CHAPTER 12

# Separations and Purifications

## Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

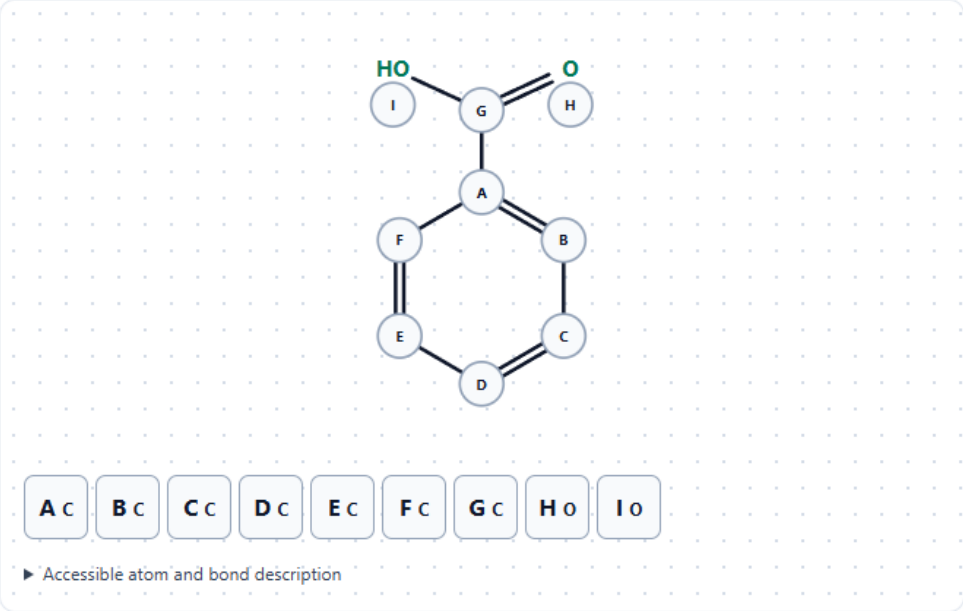
Source snapshot SHA-256: b913bafcca67f41be708f5aaaa379ad214e10c52fdb2e1f5d87b210e174790c9

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

# acid-extraction | Independent exercise

Screenshot: acid-extraction-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

## acid-extraction:stage-1 | Select the complete carboxyl group.

Answer type: atoms

## acid-extraction:stage-2 | To favor transfer of this acid into an aqueous phase, choose…

Answer type: choice

Option 0 (A): A pH well above its pKa

Option 1 (B): A pH far below its pKa in every case

## acid-extraction:stage-3 | To collect recrystallized solid from cold mother liquor, use…

Answer type: choice

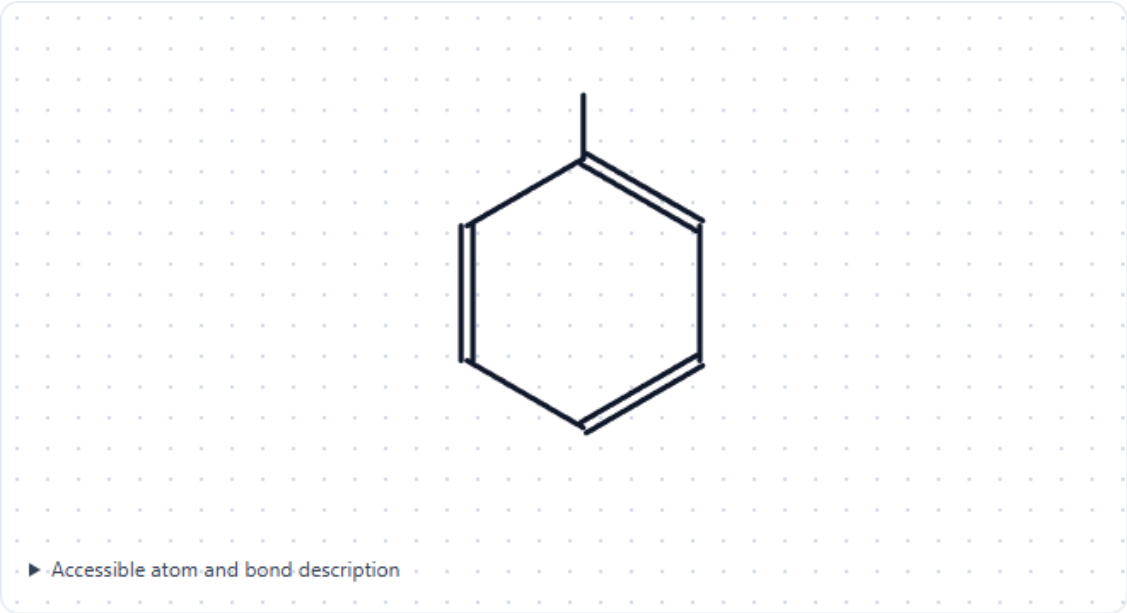
Option 0 (A): Vacuum filtration

Option 1 (B): Hot gravity filtration to keep all product dissolved

Option 2 (C): Simple distillation of the crystals

# choose-distillation | Independent exercise

Screenshot: choose-distillation-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): C; x=170.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1]]

## choose-distillation:stage-1 | A close-boiling liquid mixture is generally better separated with…

Answer type: choice

Option 0 (A): Fractional distillation

Option 1 (B): Simple filtration

## choose-distillation:stage-2 | Reducing external pressure generally makes the boiling temperature…

Answer type: choice

Option 0 (A): Lower

Option 1 (B): Higher

Option 2 (C): Exactly unchanged

## choose-distillation:stage-3 | Which can limit ordinary fractional distillation?

Answer type: choice

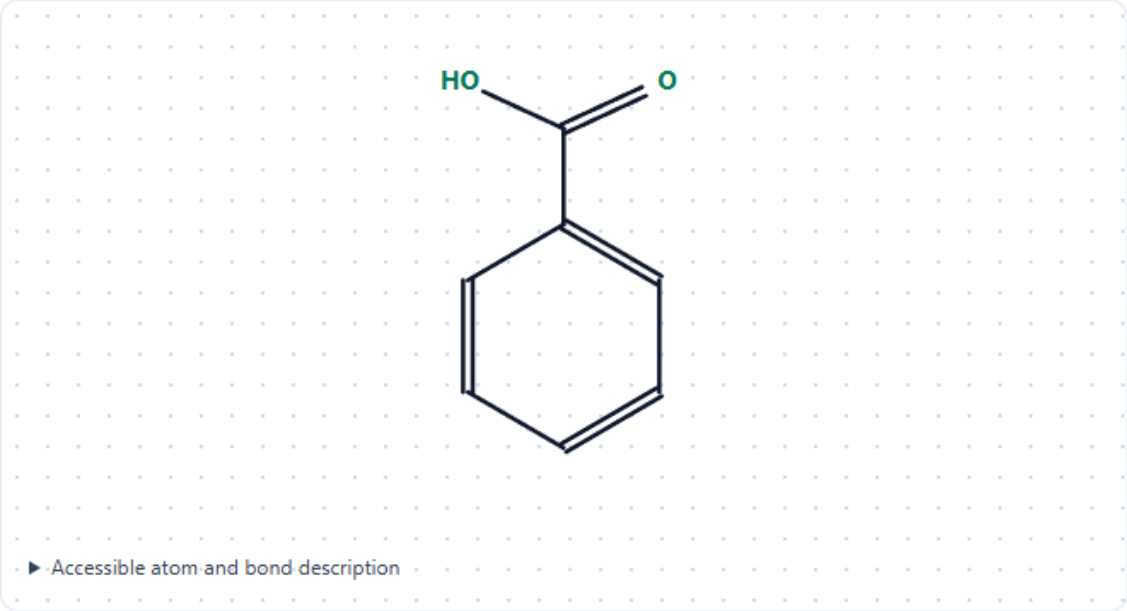
Option 0 (A): An azeotrope

Option 1 (B): The presence of any carbon atom

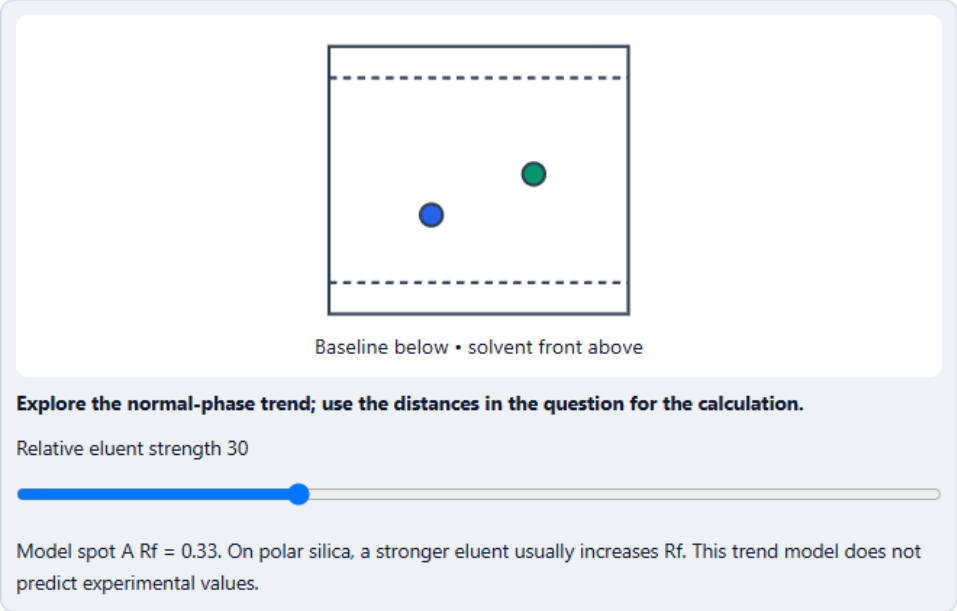
Option 2 (C): A colored liquid by itself

# rf-calculation | Independent exercise

Screenshot: rf-calculation-stage-1-molecule




Screenshot: rf-calculation-diagram




{"diagramSpecification": {"type": "tlc", "caption": "Explore the normal-phase trend; use the distances in the question for the calculation."}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

## rf-calculation:stage-1 | A spot center travels 1.8 cm and the solvent front 6.0 cm. What is Rf?

Answer type: number | unit: dimensionless

## rf-calculation:stage-2 | On normal-phase silica, a stronger eluent usually makes Rf…

Answer type: choice

Option 0 (A): Increase

Option 1 (B): Decrease in every case

## rf-calculation:stage-3 | Two compounds with the same Rf on one plate are…

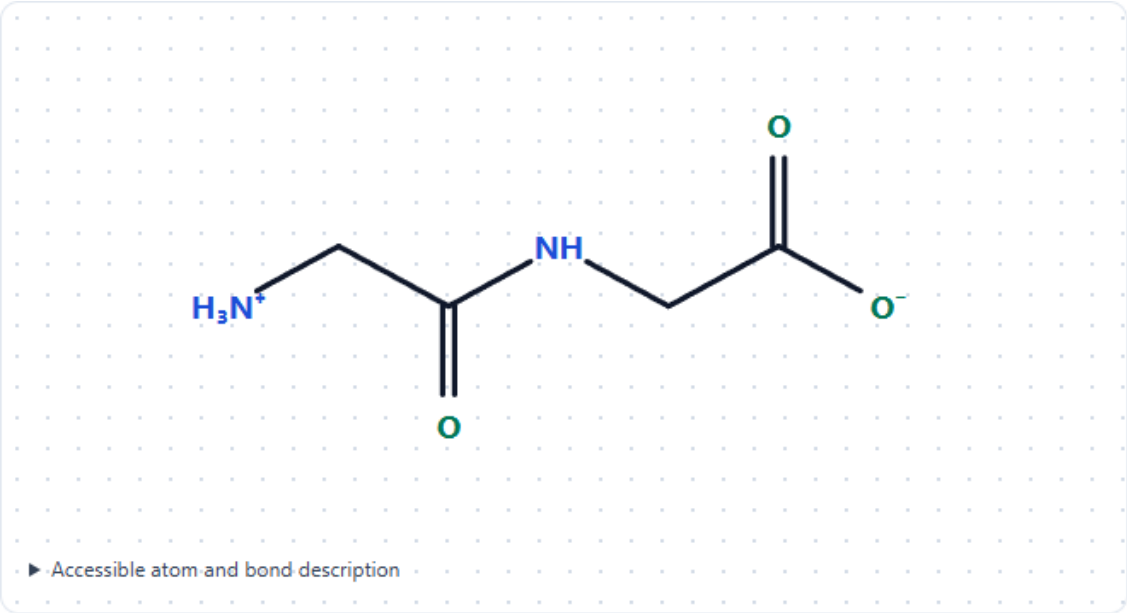
Answer type: choice

Option 0 (A): Not proven identical

Option 1 (B): Necessarily identical

# chromatography-match | Independent exercise

Screenshot: chromatography-match-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

## chromatography-match:stage-1 | A protein at pH 7 with pI 5 tends to have what net charge?

Answer type: choice

Option 0 (A): Negative

Option 1 (B): Positive

## chromatography-match:stage-2 | Which resin would electrostatically bind that protein?

Answer type: choice

Option 0 (A): Positively charged anion exchanger

Option 1 (B): Negatively charged cation exchanger

## chromatography-match:stage-3 | In ideal size exclusion, larger molecules within the useful separation range usually elute…

Answer type: choice

Option 0 (A): Earlier

Option 1 (B): Later